

Domain-Aware Speculative Decoding

WMT16 per-cluster drafters — results summary

Target: TurboSparse-Mistral-Instruct (7B) • Drafter: Lite-Mistral-150M-v2-Instruct (156M) • Loss: 0.5·CE + 0.5·KL (T=1.0) • 10 epochs • 1× RTX

Training summary

Drafter	Language	Samples	Final eval_loss	Final top-1 acc	Train time
per-cl. ru-en	Russian → English	28,500	2.363	50.04%	0.9 h
per-cl. cs-en	Czech → English	28,500	2.260	49.96%	0.8 h
per-cl. fi-en	Finnish → English	28,500	2.252	49.36%	1.0 h
per-cl. tr-en	Turkish → English	28,500	1.998	54.91%	0.9 h
per-cl. de-en	German → English	28,500	2.307	52.77%	0.9 h

Overlap area (acceptance-rate proxy)

higher is better

Drafter \ Eval cluster	ru-en	cs-en	fi-en	tr-en	de-en
per-cl. ru-en	0.4642	0.4189	0.4161	0.4703	0.4330
per-cl. cs-en	0.4149	0.4648	0.4175	0.4671	0.4307
per-cl. fi-en	0.4111	0.4158	0.4649	0.4670	0.4253
per-cl. tr-en	0.4127	0.4157	0.4184	0.5203	0.4293
per-cl. de-en	0.4158	0.4194	0.4181	0.4672	0.4689
Text Reformulation	0.4436	0.4422	0.4455	0.4994	0.4519
Base (untrained)	0.3626	0.3686	0.3733	0.4284	0.3762

Top-1 match

higher is better

Drafter \ Eval cluster	ru-en	cs-en	fi-en	tr-en	de-en
per-cl. ru-en	0.5390	0.4895	0.4752	0.5416	0.5037
per-cl. cs-en	0.5063	0.5306	0.4828	0.5382	0.5040
per-cl. fi-en	0.5033	0.4895	0.5245	0.5351	0.4988
per-cl. tr-en	0.5055	0.4890	0.4809	0.5870	0.5017
per-cl. de-en	0.5039	0.4895	0.4780	0.5388	0.5450
Text Reformulation	0.5285	0.5116	0.5115	0.5689	0.5301
Base (untrained)	0.4546	0.4395	0.4294	0.5014	0.4484

Per-cluster drafter vs. Text Reformulation baseline (overlap area)

ru-en cluster	per-cluster 0.4642 vs TextRef 0.4436	+0.0206
cs-en cluster	per-cluster 0.4648 vs TextRef 0.4422	+0.0226
fi-en cluster	per-cluster 0.4649 vs TextRef 0.4455	+0.0194
tr-en cluster	per-cluster 0.5203 vs TextRef 0.4994	+0.0209
de-en cluster	per-cluster 0.4689 vs TextRef 0.4519	+0.0170

Every per-cluster drafter outperforms the Text Reformulation baseline (11 clusters, 327K samples) on its own cluster despite using 12× less data. Deltas are ~2× larger with corrected metrics.

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Additional metrics

Top-K overlap (K=10)					higher is better
Drafter \ Eval cluster	ru-en	cs-en	fi-en	tr-en	de-en
per-cl. ru-en	0.6145	0.5592	0.5541	0.6154	0.5897
per-cl. cs-en	0.5832	0.6008	0.5525	0.6121	0.5899
per-cl. fi-en	0.5827	0.5553	0.5966	0.6132	0.5866
per-cl. tr-en	0.5846	0.5545	0.5523	0.6529	0.5874
per-cl. de-en	0.5811	0.5571	0.5516	0.6147	0.6128
Text Reformulation	0.6039	0.5790	0.5806	0.6385	0.6034
Base (untrained)	0.5526	0.5249	0.5235	0.5894	0.5504

KL divergence					lower is better
Drafter \ Eval cluster	ru-en	cs-en	fi-en	tr-en	de-en
per-cl. ru-en	6.4775	7.9709	7.6204	5.9280	7.2757
per-cl. cs-en	7.5427	6.6018	7.6342	5.9676	7.3001
per-cl. fi-en	7.5663	8.0045	6.3320	5.9183	7.3501
per-cl. tr-en	7.5161	8.0819	7.6314	4.8701	7.2688
per-cl. de-en	7.5509	8.0250	7.7069	5.9671	6.4069
Text Reformulation	6.9064	7.2823	6.8279	5.2938	6.8089
Base (untrained)	8.5757	9.1054	8.6500	6.7805	8.5054

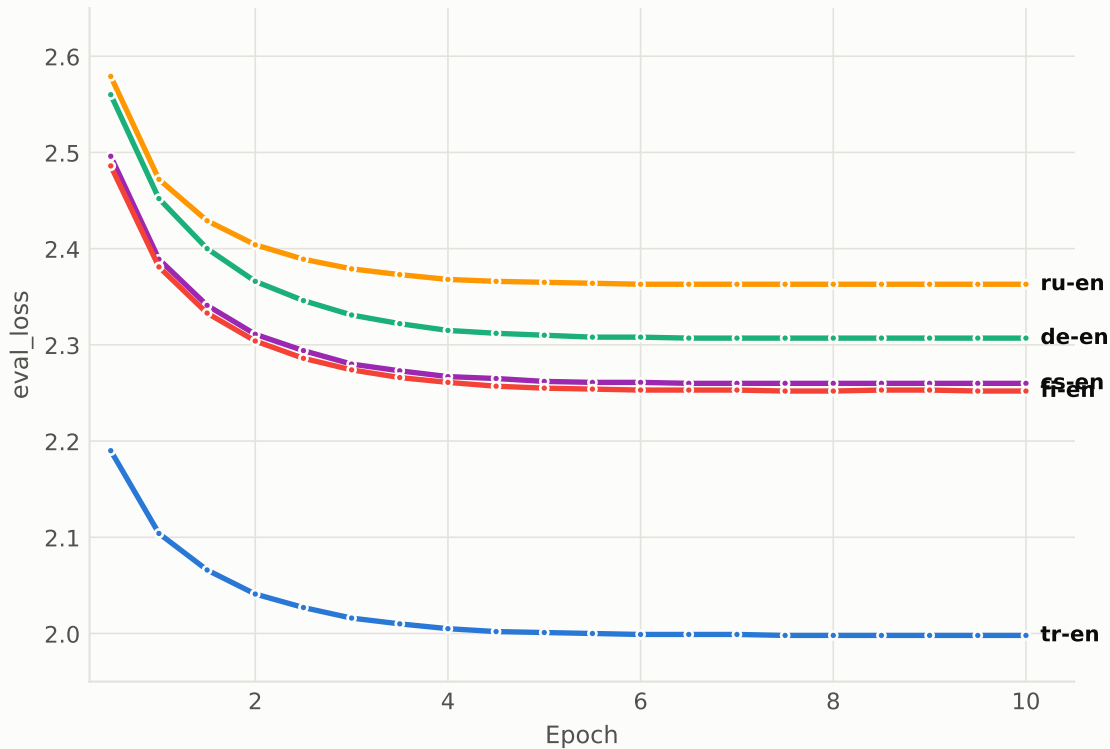
Training curves

Validation metrics, evaluated twice per epoch, 10 epochs

ru-en cs-en fi-en tr-en de-en

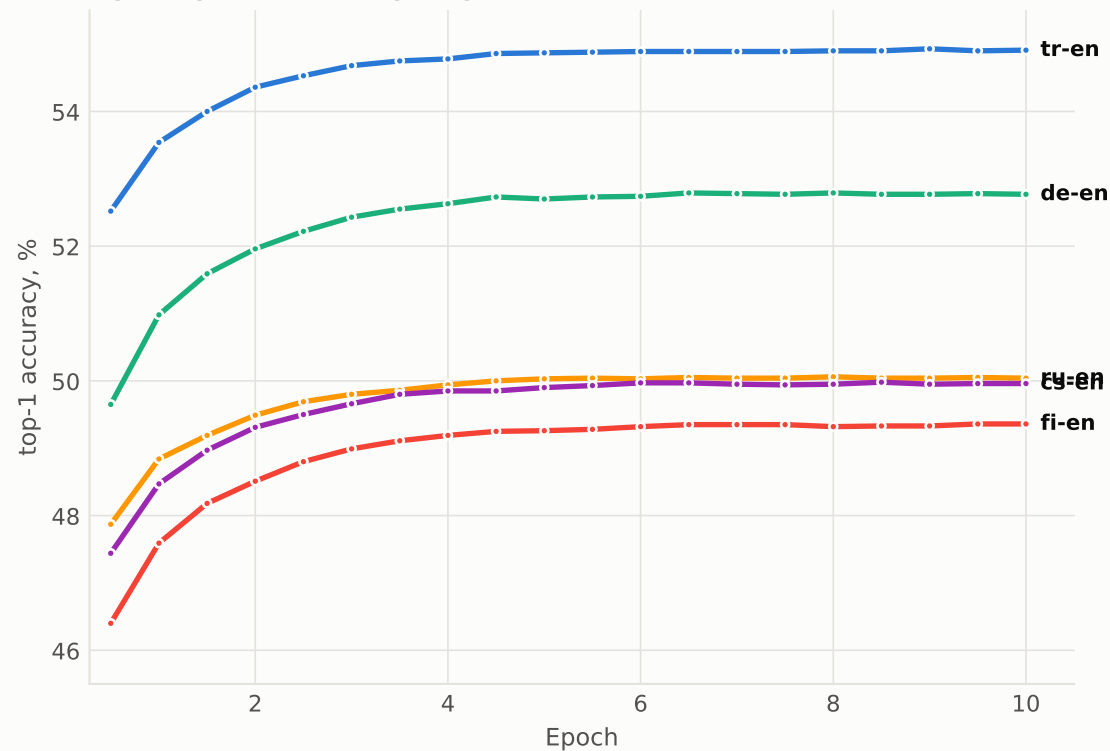
Evaluation loss

$0.5 \cdot \text{CE} + 0.5 \cdot \text{KL}$, lower is better



Top-1 accuracy

argmax agreement with target, higher is better



All five models plateau by epoch 5-6; remaining epochs yield <0.002 improvement in eval_loss.
No overfitting observed in any model (eval_loss monotonically non-increasing).